F24-189-D-Pixel

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Chapter 1

Introduction

Decentra revolutionizes the way people buy and sell goods online. It enables users to trade goods and services directly without the need of an intermediary like eBay or Amazon. It is built on the Base blockchain, offering a permissionless, transparent, anti censorship platform that eliminates middlemen, ensuring user privacy, low transaction fees, and secure trading. By using smart contracts, Decentra allows users to trade directly with each other without the restrictions and fees imposed by traditional marketplace platforms.

This report has been written for a diverse set of readers, each with unique roles in relation to our marketplace: Developers: The developers are responsible for developing the decentralized protocol from the ground up. The developers can build on the use case and architectural diagrams outlined in this document. They will benefit from the technical specifications of Decentra's smart contracts and blockchain integration, ensuring the system is robust, scalable and available for all.

Product Owners: Responsible for overseeing the overall product vision and its implementation, these stakeholders will use the document to align the project's goals with its development. They will focus on ensuring that the features and timeline meet the market's needs.

Business Strategists: They are responsible for identifying and capitalizing on market opportunities, strategists will refer to this document to understand Decentra's value proposition, competitive advantages, and user-centric features, helping them position the product effectively.

End Users (Traders and Store Owners): Individuals and businesses intending to use Decentra, either to buy and sell assets or set up online stores, will benefit from understanding how the platform reduces fees, enhances privacy, and allows decentralized trading. Hence providing more transparency and building more trust on the platform.

Security Analysts and Testers: Experts focused on assessing and pen testing the platform's security can use the document to identify key areas where vulnerabilities and bugs could arise, particularly in the smart contract interactions and P2P network. This will assist them in performing risk assessments and developing security measures.

Marketing Staff: This document provides marketing professionals with a clear understanding of Decentra's unique selling points, such as its decentralized nature, privacy protection, and cost advantages. These insights can help them craft marketing strategies aimed at attracting users and promoting the platform.

This document serves as a roadmap for all involved entities/parties, ensuring that they fully understand Decentra's goals, architecture, and development process, facilitating collaboration across different areas of expertise.

1.1 Problem Statement

The development of Decentra roots from the increasing dissatisfaction with the centralized nature of online traditional marketplace. These platforms, which dominate the e-commerce niche, impose significant restrictions on both sellers and buyers, resulting in higher listing and transaction fees, constrained trade, and immense privacy concerns. Centralized platforms typically act as middlemen, charging heavy listing and transaction fees that range from 9

Beyond the privacy and financial concerns, centralized traditional platforms also limit the users' freedom by controlling the way which the users can pay on their platform, restricting them to traditional payment methods like credit/debit cards or PayPal/Paytm. These payment methods are not only prone to high processing fees but also subject to delays, chargebacks, and frustrating disputes. The centralized control over user data, trade policies, and payment methods places an undue burden on users, who are forced to comply with the platform's requirements and face risks to their security and privacy.

In response to these challenges, Decentra offers a decentralized alternative to this whole ecosystem, designed to eliminate the need for intermediaries and empowering users with more control and freedom over their transactions. By leveraging smart contracts and blockchain technology, our platform creates a permissionless, anti censorship marketplace where buyers and sellers can interact directly with each other. The platform operates using smart contracts on the Base blockchain, which allows for automated and transparent transactions without the need for a trusted third party, hence offering users more flexibility and options for payment methods. Smart contracts provide a self-executing mechanism that ensures trade terms are met, therefore minimizing the likelihood of fraud and disputes. This completely removes the need for centralized arbitration or oversight, as all transactions are governed by pre-defined, immutable smart contract code.

A key advantage of decentralization for users are the lower transaction fees. Since there is no middleman taking a cut of each sale, sellers can list their products at competitive prices, and buyers can purchase products and services without the added burden of platform fees. Plus, our platform supports the use of various cryptocurrencies, thus giving users the flexibility to choose their preferred payment method, which can further reduce fees and enhance the speed of transactions. This approach ensures that users retain full ownership of their data and assets, as the blockchain's cryptographic foundations protect user privacy and prevent unauthorized access to sensitive information.

Another major benefit of our marketplace is the ability to resist censorship through decentralization. Unlike traditional centralized platforms that can unilaterally delist products or block transactions, our platform operates in a distributed manner, making it nearly impossible for any single entity/user to restrict or control access to the marketplace to anyone else. This application can open up new opportunities for global trade, allowing users and businesses in regions with restrictive trade policies to engage in ecommerce without fear of censorship or interference from governments or corporations. In terms of user privacy, our platform ensures that personal information is shared only when absolutely necessary, and even then, only with relevant parties. By using public key encryption and cryptographic signatures, Decentra allows users to maintain their anonymity while still engaging in secure and verifiable transactions. This stands in stark contrast to traditional centralized platforms, which often require users to provide extensive personal information that is stored on vulnerable servers, making it a prime target for attackers.

In summary, Decentra takes into account the key limitations of centralized ecommerce platforms by offering a decentralized, permissionless blockchain-based marketplace that empowers users with enhanced freedom, lower fees,greater privacy, and resistance to censorship. By removing the need for middlemen through leveraging smart contracts, Decentra provides a more secure, transparent, and efficient trading ecosystem where users can trade freely without being subject to the policies and fees of centralized authorities. This decentralized marketplace is not just a technological shift, but a necessary evolution in the way global commerce operates, putting control back in the hands of users while ensuring their privacy and security.

1.2 Scope

Our scope for the application Decentra is to develop two things:

1. A *Decentra Protocol* that is a collection of smart contracts that allow users to trade in a decentralized manner. Anybody is free to build a Client that interacts with this protocol to enable trading. Other people might want to build a Client for various reasons For example: Our hosted Client*(Decentre Web Application)* may have to censor selling of *specific goods* due to legal obligations. Another person can host

a web application interacting with the protocol that doesn't censor these *specific goods*. The protocol remains *uncensored* and completely *decentralized* at all times. The collection of smart contracts*(Decentra Protocol)* is similar to what web developers would traditionally call an API and will be documented as such.

2. A *Decentra Web Application(Client)* that allows end users to interact with the smart contracts. The interactions with smart contracts will be heavily abstracted so there is no learning curve for the end user. This also consists of any indexers or other auxillary modules that would be required to build a user-friendly web application. This will be extensively documented to ensure anybody else wanting to build a Client to interact with the smart contracts can learn by example.

One of the core functionalities of Decentra is the ability to perform secure and transparent transactions using cryptocurrencies, with support for any ERC20 tokens. This offers users the freedom to choose from a diverse range of digital currencies for payments, breaking the dependence on traditional banking systems and debit/credit card payments. By utilizing smart contracts, the platform automates transaction processes, including payment handling, escrow services, and dispute resolution, ensuring fairness and transparency without the need for centralized arbitration.

An important feature of Decentra is its Trade Protocol, which governs the terms and conditions of transactions between buyers and sellers. The protocol supports multiple types of payment methods, including direct payments and double deposit escrow systems, where both parties must agree from their sides before funds are released. For additional security, the platform incorporates Notary Escrow, where a third party notary can be chosen to arbitrate transactions in the event of a dispute. This system ensures that users have access to fair and unbiased resolution mechanisms without compromising decentralization. Furthermore, all transaction metadata, including trade agreements, assets or services being traded, and personal information of users, are all securely stored within the smart contracts, ensuring transparency and securing information of our users.

To enhance the decentralized nature of the marketplace, we will use IPFS (InterPlanetary File System) to store product listing images, reviews, and other relevant data in a distributed manner. By hosting this data on IPFS, the platform ensures that information is safe from tampering and not reliant on any single centralized server. This helps increase the reliability of the platform and ensures that users can access product information anytime increasing the availability aspect of our platform, even in the case traditional servers experience downtime or technical issues. The use of IPFS further reinforces our platform's commitment to creating a fully decentralized environment where users are not reliant on any single point of failure.

The Decentra Client Application, a web-based application, allows users to interact with the platform in various roles, including merchants, buyers, and notaries. Merchants can set up online stores, list products, and manage sales, while buyers can search for products, make purchases, and leave reviews. The application is equipped with a user-friendly interface that simplifies interactions with the decentralized backend, enabling users with minimal technical knowledge to navigate the platform easily. It also includes a ratings and reviews system that stores feedback about merchants and products on IPFS, ensuring that reviews are permanent and cannot be tampered with. This promotes trust and accountability on the platform, as buyers can make informed decisions based on the experiences of others.

Moreover, the scope also includes the implementation of a Notaries and Voting Pools system. This feature allows users to choose notaries to act as arbiters in the case of disputes, or rely on a pool of selected judges to provide arbitration services. Voting polls help improve fairness and transparency by involving multiple arbiters/judges in the decision-making process, further ensuring fairness and transparency in trade. The system is made in a way to be highly flexible, allowing business owners and buyers to either manually select their preferred notaries or rely on the platform's automated selection process, which is defined in our smart contracts i.e maintaining transparency throughout the process.

The project also includes an escrow system backed by multisignature smart contracts to ensure that payments are only released when both buyers and sellers have approved the transaction. This approach greatly reduces the risks attached with online purchasing, such as fraud or non delivery of products or services. Furthermore, our platform incorporates decentralized search tools, enabling users to efficiently browse and find the products they need without relying on centralized algorithms or databases.

In summary, the scope of this marketplace covers the development of a decentralized marketplace protocol and a client to create a secure, transparent, permission-less ,and an open trading ecosystem.

Its main functionalities include:

1. Cryptocurrency Based Transactions
2. Escrow Functionality
3. Dispute Resolution Mechanisms
4. Tamper-Proof Review System

All designed to ensure a seamless and trustworthy trading experience for global users. The project's decentralized architecture aims to remove the traditional constraints of e-commerce, offering users a platform that is censorship-resistant, cost-effective, and fully under their control.

1.3 Modules

1.3.1 Trade Protocol

The Trade Protocol governs the transactions on our platform. It has smart contracts that setup trade terms between users, manage transactions, and handle disputes between the peers. The system supports multiple payment methods, including direct payments and escrow using smart contracts.

Features:

1. **Base Chain Payments and Escrow**

    - **Direct Payment**: A direct trade between a buyer and seller, used when buyer trusts the seller.
    - **Double-deposit Escrow**: 2 of 2 multisignature smart contract. An escrow where both the buyer and seller trust each other. Fund will stay in the smart contract if any of the parties refuse signature.
    - **Notary Escrow**: 2-of-3 or n-of-m multisignature smart contract with notariy. The buyer and seller agree on a arbiter prior to beginning the trade. The arbiter will have to sign off on the transaction.
    - **Smart Contracts**
        - **Metadata Module**: Defines trade type and validity period.
        - **ID Module**: Contains network ID, Account pubkey, PGP pubkey, and other identifiers.
        - **Trade Module**: Information of the goods or services being traded.

2. **IPFS for Hosting Product Listing Images and Reviews** Files are stored in a way that prevents tampering and ensures permanence, leveraging IPFS's content-addressable system.

    - **Product Listing Images:** Store high-resolution images of products on IPFS to ensure they are always accessible without relying on a single point of failure. This enables faster load times and reduces reliance on centralized servers.
    - **Product Reviews:** Host user reviews and feedback on IPFS, ensuring they are permanently recorded and tamper-proof. Each review is stored as a separate file, accessible via unique content hashes.

1.3.2 Decentra Client Application (Web App)

The Decentra client application is central to user interaction with the marketplace. It allows users to take roles such as merchant, buyer, notary, or arbiter, with diverse UI/UX depending on the type of trade. The application includes:

1. **User Role Management**
    - **Merchant**: Allows for setting up store , product listing, and managing sales.
    - **Buyer**: Facilitates searching, purchasing, and payment management.
    - **Notary**: Provides arbitration system and validates reviews.
2. **Decentralized Network Interaction**
    - Connects with the smart contracts on deployed on Base(Chain).
    - Connects with IPFS to show images.
3. **Ratings and Reviews System**
    - **Ratings**: Scores for merchant behavior, item quality, and shipping time.
    - **Reviews**: Written summaries of trade experiences.
    - **Considerations**: Mitigation of Sybil attacks, persistent and immutable storage.

1.3.3 Notaries and Voting Pools

- **Arbiters**
- **Function:** Provide arbitration and review validation services.
- **Accreditation**: Carry over off chain trust.
- **Choosing Notaries**: Users can manually select or rely on notaries they like provided both parties agree.

1. **Voting Pools**
    - **Purpose**: Minimize collusion risk by involving multiple arbiters.
    - **Pool Formation**: Includes notaries chosen by merchant and buyer, plus random selections.
    - **Digital Rulings**: Notary pool receives a fee for the moderating the trade.

1.4 User Classes and Characteristics

User Classes and Characteristics

User class	Description
Individual Sellers	Typically small business owners, artisans, or hobbyists. Interested in selling goods and services without incurring high fees. Prefer platforms that offer autonomy and control over their listings. Value privacy and wish to manage what personal information is shared.
Buyers	Buyers using Decentra are individuals in search of a diverse array of products and services, drawn to the platform by its competitive pricing and extensive selection. They prioritize a shopping experience that is both secure and private, with transparency in transactions being a key factor. The absence of additional fees during transactions is particularly appealing to these users, as they seek a frictionless purchasing process without hidden costs.
Marketplace Operators	Are focused on establishing niche marketplaces within the Decentra ecosystem. They require robust tools for managing multiple vendor accounts and overseeing transactions while ensuring a user-friendly interface for end-users. These operators are keenly interested in analytics and reporting features that provide insights into marketplace performance, helping them optimize their operational strategies.
High-Trust Individuals	Can provide valuable escrow services, acting as intermediaries to facilitate secure and reliable transactions between buyers and sellers. These escrow service providers play a critical role in building trust within the decentralized marketplace by ensuring that both parties fulfill their obligations before the exchange of goods, services, or digital assets is completed.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case

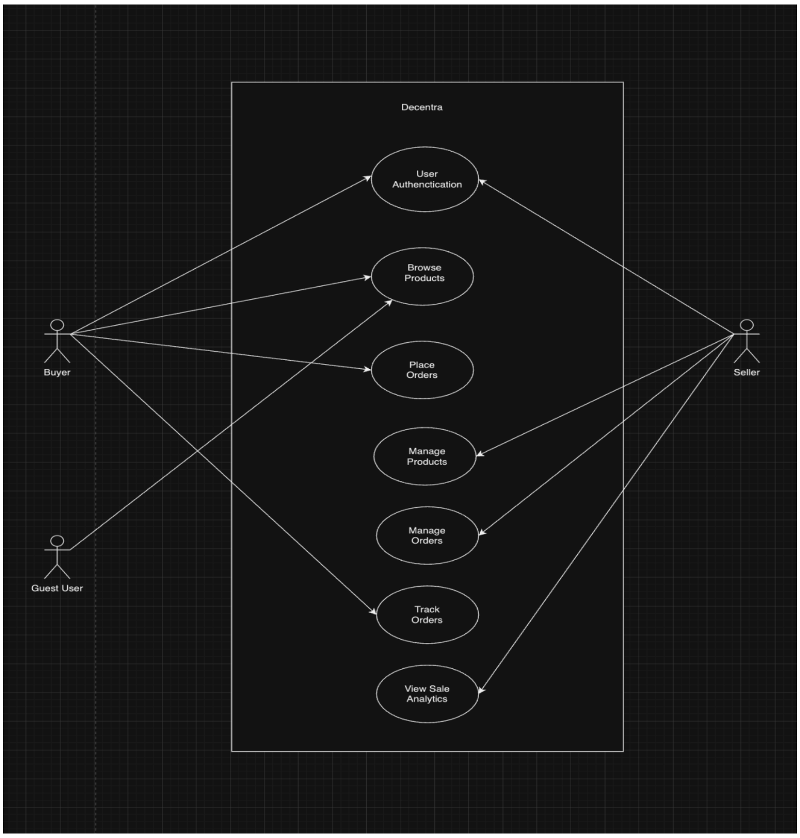

Figure 2.1: Use Case Diagram for Decentra

2.1.1 Actors

- **Buyer:** A user who has registered and logged into the platform.
- **Seller:** A user who lists and manages products for sale on the platform.
- **Guest User:** A user who browses the marketplace without logging in or registering.

2.1.2 Browsing Products

Actors: Guest User, Buyer
Description: The user (guest or buyer) can browse products listed in the marketplace. Guest users can view the products but need to sign up or log in to perform further actions like adding to the cart or purchasing.
Preconditions: The platform must have products listed by sellers.

Main Flow:

1. The user navigates to the homepage or search page.
2. The user can filter products by category, price, or other parameters.
3. The system displays product details such as price, seller, and description.
4. If the user is a buyer, they can add products to the cart.
5. Guest users are prompted to log in or sign up to proceed to checkout.

Postconditions: The user views product details, and buyers can add items to their cart.

2.1.3 Add to Cart

Actor: Buyer
Description: A logged-in buyer can add selected products to their cart for future checkout.
Preconditions: The buyer must be logged in. Products must be available in the marketplace.

Main Flow:

1. The buyer selects a product to view.

2. The system displays the product's details.
3. The buyer clicks the "Add to Cart" button.
4. The system adds the product to the buyer's cart and confirms the action.

Postconditions: The product is added to the buyer's cart, ready for checkout.

2.1.4 Place Order

Actor: Buyer
Description: The buyer can place an order for products added to the cart.
Preconditions: The buyer must have products in their cart and payment information ready.

Main Flow:

1. The buyer proceeds to checkout.
2. The system prompts the buyer to enter shipping and payment details.
3. The buyer reviews the order summary.
4. The buyer confirms the order, and the system processes the payment.

Postconditions: The order is placed, and the buyer receives an order confirmation.

2.1.5 Manage Products

Actor: Seller
Description: Sellers can add, update, or remove products in the marketplace.
Preconditions: The seller must be logged in.

Main Flow:

1. The seller navigates to the seller dashboard.
2. The seller can click on "Add New Product" to list a new item for sale.
3. The system prompts the seller to enter product details, including images and price.
4. Sellers can also update or remove existing product listings from the dashboard.

Postconditions: New products are added to the marketplace or existing products are updated/removed.

2.1.6 Manage Orders

Actor: Seller
Description: The seller manages incoming orders placed by buyers, updates the order status, and processes shipping.
Preconditions: The seller has products listed in the marketplace and received orders.

Main Flow:

1. The seller navigates to the order management section.
2. The seller views new orders and selects an order to update.
3. The system displays order details, and the seller can update the status (e.g., "Shipped," "Delivered").

Postconditions: The order status is updated, and the buyer is notified.

2.1.7 Tracking Orders

Actor: Buyer
Description: The buyer can track the status of orders they have placed.
Preconditions: The buyer must have placed an order and received a tracking number or status.

Main Flow:

1. The buyer navigates to the "My Orders" section in their account.
2. The system displays all past orders with current statuses (e.g., "Processing," "Shipped," "Delivered").
3. The buyer selects an order to view detailed tracking information (e.g., courier details, estimated delivery).

Postconditions: The buyer can see the real-time status of their order and contact support if necessary.

2.1.8 User Authentication

Actors: Buyer, Seller
Description: Both buyers and sellers must authenticate (log in or sign up) to access platform features like placing an order or managing products.
Preconditions: The user has an account or must create one.

Main Flow:

1. The user clicks on the login/signup button.
2. The system prompts the user to enter email and password or sign up using a registration form.
3. The system authenticates the user and redirects them to the respective dashboard (buyer or seller).

Postconditions: The user is authenticated and can now access buyer/seller-specific features.

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

2.2.1 Trade Protocol

Following are the functional requirements for the Trade Protocol module:

1. The system must allow buyers to make direct payments to sellers without escrow when trust is established.
2. The system must implement a 2-of-2 multisignature smart contract for double-deposit escrow, requiring signatures from both parties for transaction completion.

3. The system must support a 2-of-3 or n-of-m multisignature smart contract involving a designated notary, requiring their signature for transaction completion.
4. The system must allow for the definition of trade type and validity period, ensuring trades comply with these terms.
5. The system must maintain identifiers such as network ID, account public key, and PGP public key for each transaction.
6. The system must store information related to the goods or services being traded.
7. The system must provide a mechanism for handling disputes between peers.
8. The system must support various payment methods, including direct payments and escrow transactions.

2.2.2 IPFS for Hosting Product Listing Images and Reviews

Following are the functional requirements for the IPFS module:

1. The system must store high-resolution images of products on IPFS.
2. The system must host user reviews on IPFS, with each review stored as a separate file accessible via unique content hashes.
3. The system must leverage IPFS's content-addressable system for retrieving product images and reviews.

2.2.3 Decentra Client Application (Web App)

Following are the functional requirements for the Decentra Client Application module:

1. The application must allow users to select roles (merchant, buyer, notary, arbiter) with tailored features for each role.
2. Merchants must be able to set up stores, list products, and manage sales.
3. Buyers must be able to search, purchase, and manage payment transactions.
4. Notaries must provide arbitration services and validate user reviews.
5. The application must connect with smart contracts deployed on the Base chain and access images hosted on IPFS.
6. Users must be able to provide ratings and written reviews for merchants and products.

2.2.4 Notaries and Voting Pools

Following are the functional requirements for the Notaries and Voting Pools module:

1. The system must allow notaries to provide arbitration and validation services for trades.
2. Users must be able to manually select notaries or rely on suggestions, ensuring mutual agreement between parties.
3. The system must enable the formation of voting pools to minimize collusion risks during arbitration.
4. The notary pool must be able to issue digital rulings and receive fees for moderating trades.

2.3 Non-Functional Requirements

2.3.1 Reliability

The decentralized marketplace platform must have a high level of reliability, ensuring minimal downtime. All functions of the system should be deterministic. In case of any failures, the system should ensure that data is not lost and users are informed of the issue. Robust error detection mechanisms must be in place to identify faults in real-time, while automated recovery strategies should restore normal operation within a specified time, preferably within one hour of a detected failure.

2.3.2 Usability

The platform must be easy to use, catering to both novice and experienced users. Since the platform relies on decentralized wallet authentication, users should be able to create accounts, perform transactions, and manage their listings seamlessly. The design should be intuitive, ensuring users can accomplish tasks with minimal effort, without requiring technical knowledge of blockchain or smart contracts. It is critical that the interface be responsive and provide appropriate feedback in case of errors. Additionally, the system should prevent common user mistakes and offer suggestions to rectify issues when they arise. To enhance accessibility, the platform must comply with established guidelines, ensuring users with disabilities can interact with the platform effectively. For example, wallet authentication and order placement should not require more than a few straightforward steps.

2.3.3 Performance

Performance requirements for the decentralized platform must be optimized to handle high user traffic and transaction loads. Specific performance goals should include the ability to process transactions such as purchasing or listing items within a few seconds, ideally no more than 5 seconds. Search operations should provide results in under 1 second for most users. The platform should be designed to scale, handling up to 10,000 concurrent users without a noticeable decline in performance. Smart contracts, the core mechanism for transaction validation, should also be optimized for speed, ensuring that blockchain confirmations do not slow down the user experience, particularly during peak traffic times.

2.3.4 Security

The security of the decentralized platform is paramount, especially as it handles sensitive financial transactions. All user interactions, including wallet-based account creation and authentication, must be secure and immune to external threats such as phishing or man-in-the-middle attacks. Given the decentralized nature of the platform, there should be no centralized recovery options, meaning users must securely manage their private keys to access their accounts. The platform should encrypt all sensitive data, both in transit and at rest, utilizing encryption standards such as AES-256. Additionally, the system must be resilient against attacks, including DDoS attacks, ensuring continuous availability even under malicious pressure. The system should also provide robust mechanisms to prevent unauthorized access to smart contracts, and all critical operations must be logged for auditing and tracking purposes.

2.4 Domain Model

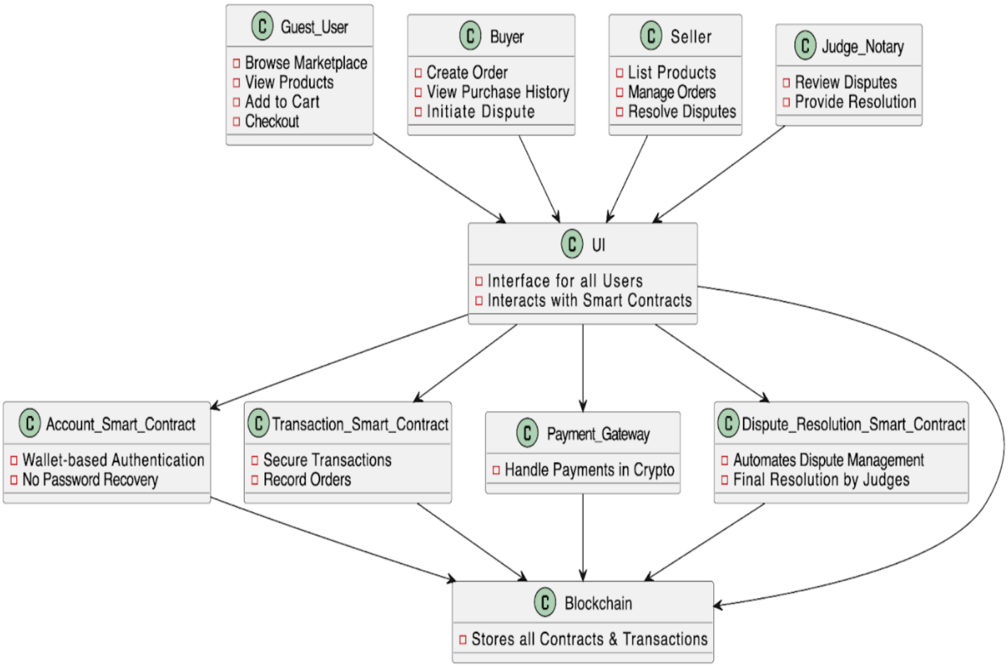

Figure 2.2: Domain Model of Decentra

2.4.1 Description

This diagram illustrates the structure of a decentralized marketplace. It highlights the interaction between different actors such as Guest Users, Buyers, Sellers, and Judge/Notaries, all interfacing with the system via a unified UI that interacts with several Smart Contracts. The Account Smart Contract manages wallet-based authentication without password recovery, ensuring security and decentralization. Transaction Smart Contracts handle secure order transactions. Payments are processed through a Payment Gateway using cryptocurrency. Dispute Resolution Smart Contracts automate dispute management, with Judges/Notaries providing final resolution. All transactions and contracts are securely stored on the Blockchain.

Chapter 3

System Overview

3.1 Architectural Design

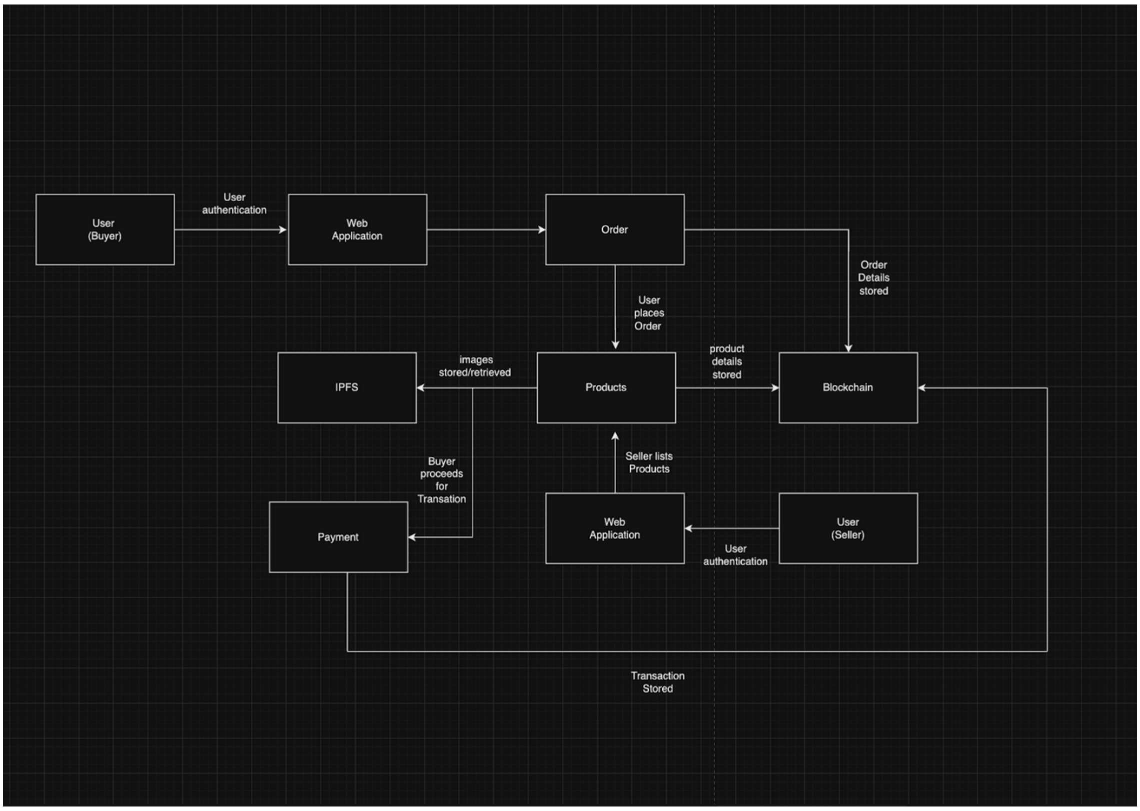

Figure 3.1: Architectural Diagram

This architecture outlines the flow between different components involved in the platform. Users, either Buyers or Sellers, first authenticate via the Web Application. Sellers list products, which are stored in the Products system along with associated images, which are handled via IPFS (InterPlanetary File System) for decentralized storage. Buyers place orders through the Web Application, and the order details are stored on the Blockchain for immutability and transparency. Buyers proceed to payment, which is processed via the Payment system using cryptocurrency, with transaction details also recorded on the Blockchain. The diagram highlights how the blockchain secures transactions, ensuring decentralization and trust between parties.

3.2 Design Models

3.2.1 Activity Diagram

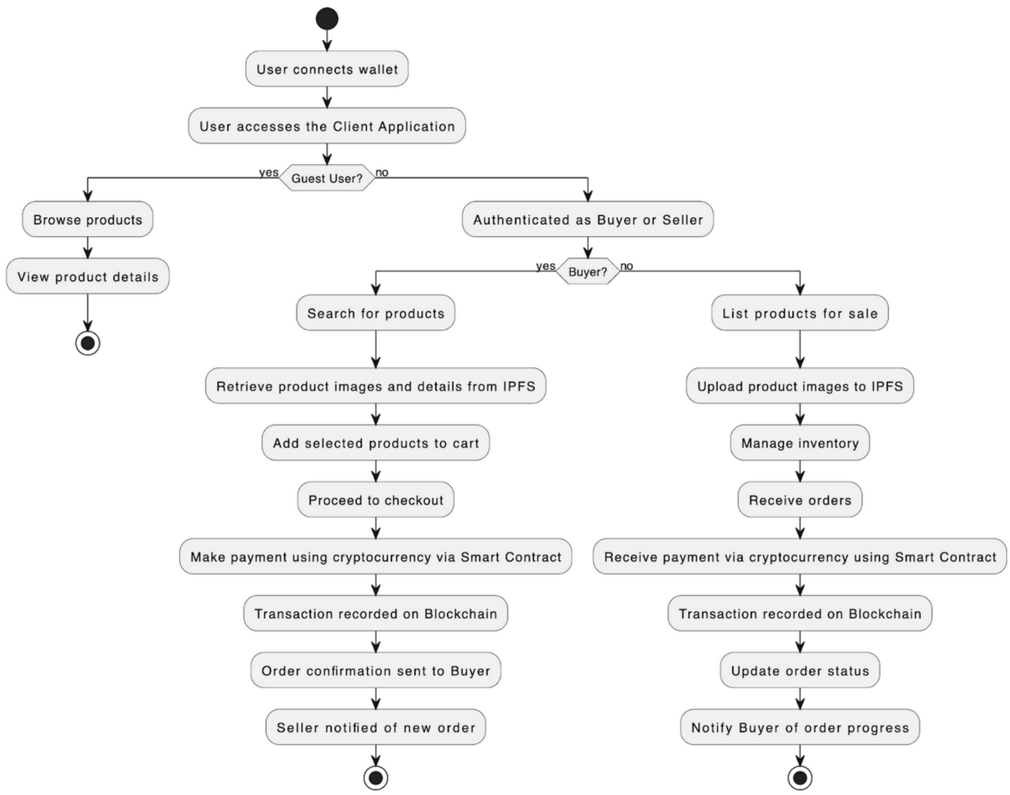

Figure 3.2: Activity Diagram

This activity diagram outlines the flow of interactions in the Dectra decentralized marketplace from the perspective of both buyers and sellers. It begins with users connecting their wallets and accessing the client application. Users can either proceed as Guest Users to browse products and view details or authenticate as Buyers or Sellers.

If a user is a Buyer, they can search for products, retrieve images from IPFS, add products to the cart, and proceed to checkout. Payment is made using cryptocurrency through a Smart Contract, which triggers the transaction to be recorded on the blockchain. Upon payment, an order confirmation is sent to the buyer, and the seller is notified of the new order.

If a user is a Seller, they list products for sale, upload images to IPFS, manage inventory, and receive orders. Payment is received via cryptocurrency through the Smart Contract, and the transaction is similarly recorded on the blockchain. After receiving payment, the seller updates the order status and notifies the buyer of the progress. This diagram effectively shows how buyers and sellers interact with the platform while ensuring decentralization and security via blockchain and IPFS.

3.2.2 Sequence Diagram

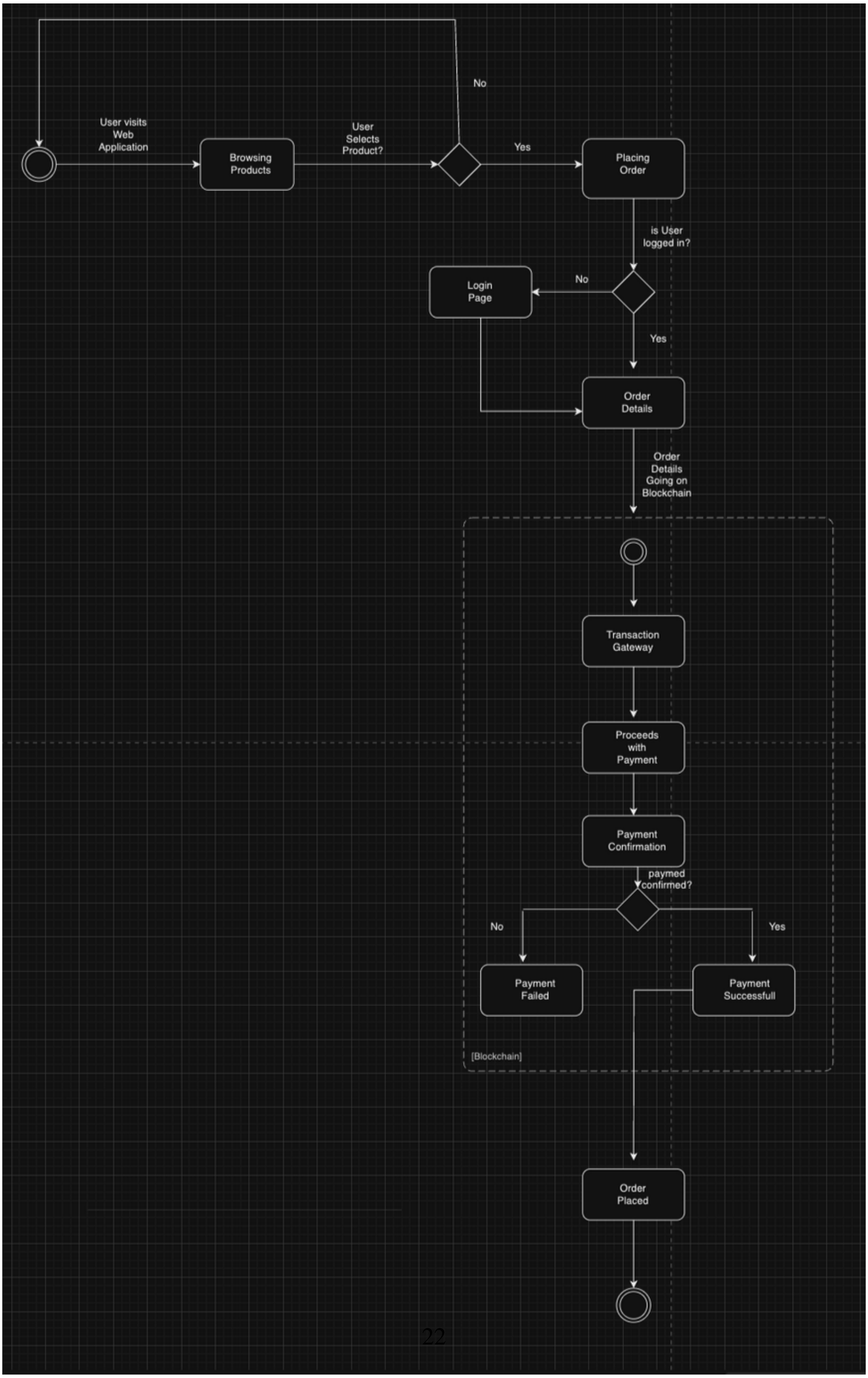

Figure 3.3: Sequence Diagram

This sequence diagram illustrates the flow of a user interacting with a decentralized marketplace. The process begins when the user visits the web application and browses products. If the user selects a product, they proceed to place an order. If not logged in, the user is redirected to a login page before continuing. Once logged in, order details are submitted to the blockchain, followed by transaction processing through a payment gateway. After payment confirmation, the system checks if the payment was successful. If successful, the order is placed; otherwise, the payment fails, ending the process.

3.2.3 Class Diagram

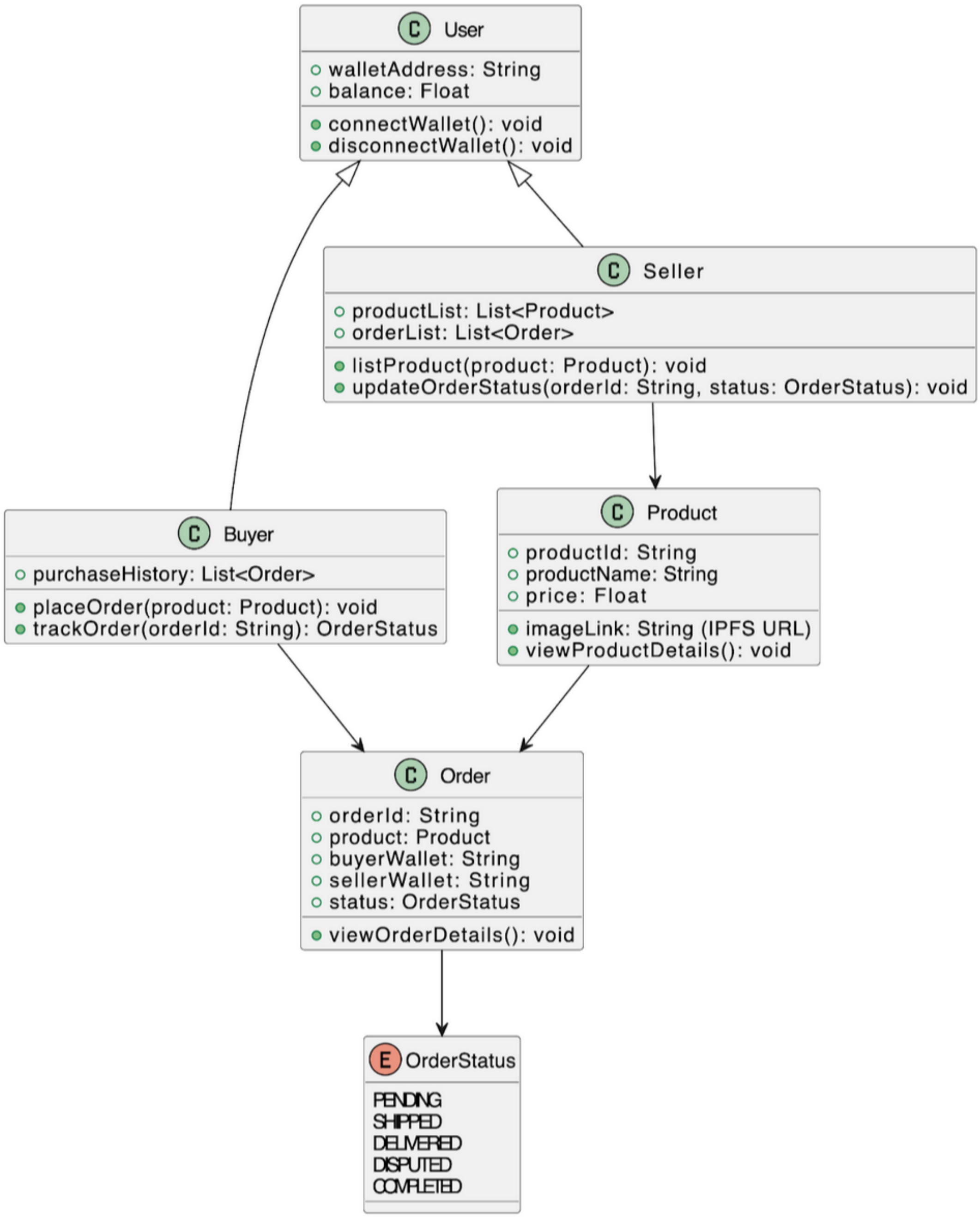

Figure 3.4: Class Diagram

The class diagram for the Decentra decentralized marketplace illustrates the primary components and their relationships within the system. The key entities are:

1. **Guest User**: Represents unregistered users who can browse the marketplace, view products, add items to the cart, and proceed to checkout.

2. **Buyer**: Extends from the general user. Buyers can create orders, view purchase history, and initiate disputes if necessary.

3. **Seller**: Responsible for listing products, managing orders, and resolving disputes. Sellers interact with the platform to manage their inventory and sales.

4. **Judge Notary**: Handles the resolution of disputes. Once a dispute is initiated, the Judge Notary reviews the case and provides a final resolution.

5. **UI**: The user interface that acts as a bridge between users and the blockchain-based smart contracts. It ensures that buyers, sellers, and guest users can interact seamlessly with the platform's functionalities.

6. **Account Smart Contract**: Manages user authentication through wallet-based authentication, ensuring privacy and security. There is no password recovery due to the decentralized nature of the system.

7. **Transaction Smart Contract**: Secures transactions and records orders. It plays a central role in managing the financial transactions between buyers and sellers.

8. **Payment Gateway**: Facilitates cryptocurrency payments for both buyers and sellers, ensuring that transactions are securely processed via the blockchain.

9. **Dispute Resolution Smart Contract**: Automates the dispute management process, allowing for fair and transparent resolutions that are finalized by the Judge Notary.

10. **Blockchain**: Serves as the backend, storing all contracts and transactions securely. It ensures transparency and immutability across all transactions and actions taken by users.

In this design, anonymity is a key feature, as minimal data from both buyers and sellers is collected. Most interactions, including account creation and payment, are handled through decentralized smart contracts, ensuring that users retain control over their information while the platform remains secure and transparent.

3.3 Data Design

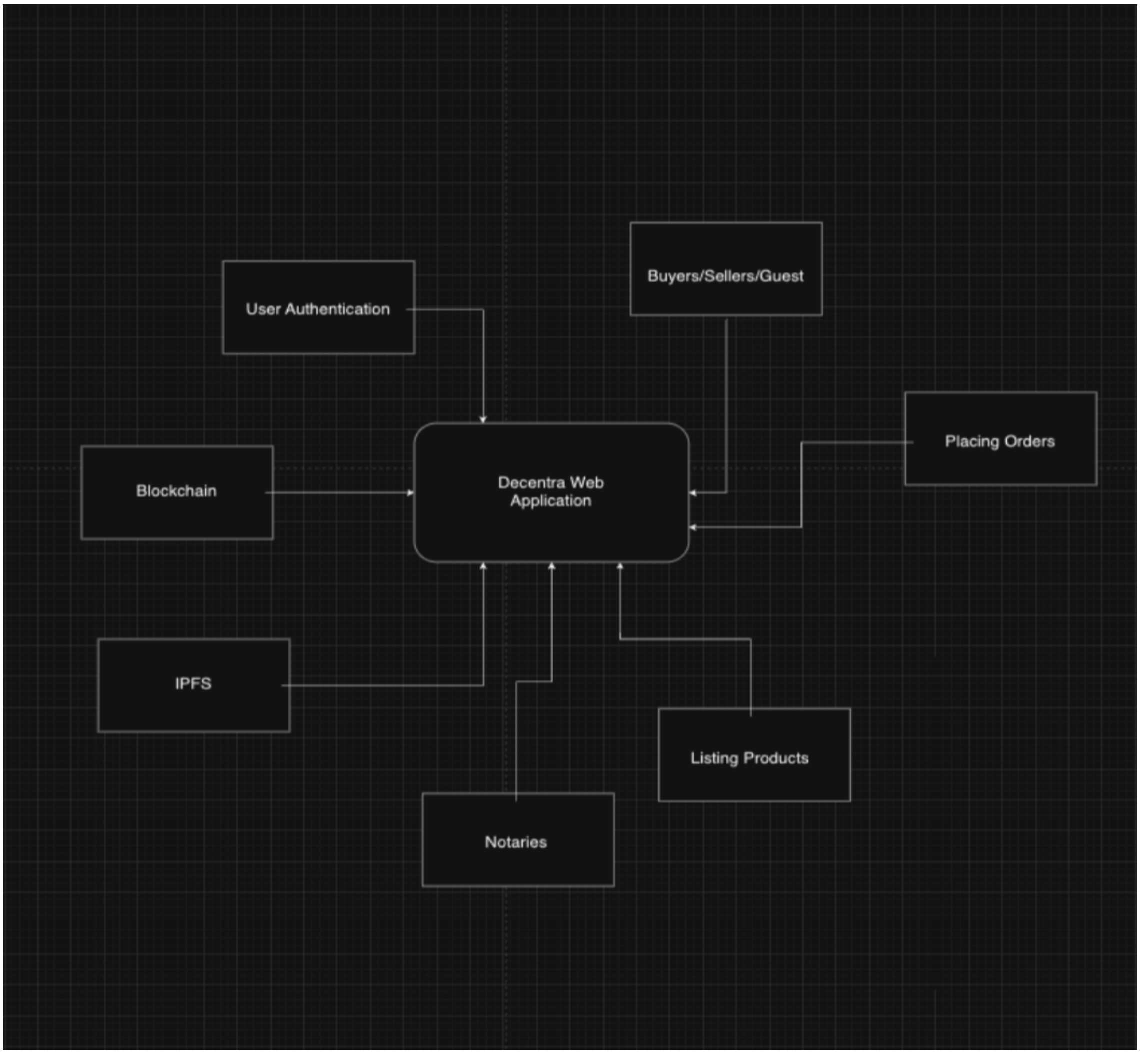

Figure 3.5: Data Flow Diagram

This Level 0 Data Flow Diagram (DFD) represents the overall structure of a decentralized marketplace web application. It identifies key components and their interactions with the central "Decentra Web Application."

1. Buyers/Sellers/Guest: External users who interact with the system. Buyers place orders, sellers list products, and guests may browse products or register. 2. User Authentication: This process ensures that only registered users can perform specific actions such as placing orders or listing products. 3. Placing Orders: Buyers can place orders for products listed on the platform through the web application. 4. Listing Products: Sellers use this feature to list products on the platform, making them available for purchase. 5.

Blockchain: The application communicates with a blockchain to store immutable transaction records and order details, ensuring transparency and security. 6. IPFS (InterPlanetary File System): This decentralized storage system is likely used to store product images, descriptions, and other large files securely and in a decentralized manner. 7. Notaries: Used for verifying transactions or product authenticity through a decentralized validation system.

Each component interacts with the central web application, forming a complete decentralized e-commerce ecosystem.

In our decentralized marketplace project, the data design primarily revolves around two key storage and processing technologies: IPFS (InterPlanetary File System) and Blockchain.

• IPFS is used to store and retrieve the large, decentralized files such as product images and descriptions. When a seller lists a product, the images and relevant metadata are uploaded to IPFS. Each file uploaded is given a unique content identifier (CID), ensuring the data is immutable and easily retrievable by both buyers and sellers. The decentralized nature of IPFS ensures that files are not hosted on a single server but spread across multiple nodes, providing redundancy and improved access speeds while maintaining data integrity. • Blockchain handles all transactions and stores the corresponding transactional data. This includes buyer and seller interactions, order creation, payments, and dispute resolution. Each transaction is securely recorded on the blockchain as a block that includes the transaction details, timestamps, and the involved parties' wallet addresses. The blockchain ensures that all order histories, payments, and contracts are immutable, transparent, and verifiable by all users without the need for a central authority.

Together, IPFS and Blockchain work in tandem to support the decentralized nature of the platform, ensuring data security, transparency, and persistence in a trustless environment. The product details stored in IPFS are linked to their corresponding transactional records on the blockchain, creating a seamless flow of data between these two components.

Bibliography